

For each of these relations on the set $\{l, m, n, o\}$, determine which relations are reflexive, symmetric, antisymmetric, and transitive. Remember that a relation can have multiple properties. Put a check mark on the property or properties that apply to each of the relations. Your answer should not exceed the correct number of properties for each relation, thus you cannot simply check all properties.

- $A = \{(m, m), (m, n), (m, o), (n, m), (n, n), (n, o)\}$
- $B = \{(l, l), (l, m), (m, l), (m, m), (n, n), (o, o)\}$
- $C = \{(m, o), (o, m)\}$
- $D = \{(l, m), (m, n), (n, o)\}$
- $E = \{(l, l), (m, m), (n, n), (o, o)\}$
- $F = \{(l, n), (l, o), (m, n), (m, o), (n, l), (n, o)\}$
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Relation	Reflexive	Symmetric	Antisymmetric	Transitive
$A = \{(m, m), (m, n), (m, o), (n, m), (n, n), (n, o)\}$	✗ (l, o missing)	✗ (m, n exists but n, m is missing)	✗ (n, m and m, n exist together)	✗ (m, n and (n, o) exist, but (m, o) is missing)
$B = \{(l, l), (l, m), (m, l), (m, m), (n, n), (o, o)\}$	✓ (contains all (x, x))	✓ ((l, m) and (m, l) exist together)	✗ (Not antisymmetric since (l, m) and (m, l) exist with $l \neq m$)	✗ ($l \rightarrow m$ and $m \rightarrow l$ do not imply $l \rightarrow l$)
$C = \{(m, o), (o, m)\}$	✗ (missing (x, x) for all elements)	✓ ((m, o) and (o, m) exist together)	✗ (Not antisymmetric since (m, o) and (o, m) exist with $m \neq o$)	✗ (No $(a, b) \rightarrow (b, c) \rightarrow (a, c)$ structure)
$D = \{(l, m), (m, n), (n, o)\}$	✗ (No (x, x) pairs)	✗ (l, m exists but not m, l)	✓ (No (a, b) and (b, a) exist together unless $a = b$)	✗ (l, m and (m, n) exist, but (l, n) is missing)
$E = \{(l, l), (m, m), (n, n), (o, o)\}$	✓ (contains all (x, x))	✓ (all pairs are (x, x) , so symmetry holds)	✓ (no distinct (a, b) and (b, a) exist together)	✓ (trivially transitive since it only contains self-pairs)
$F = \{(l, n), (l, o), (m, n), (n, l), (n, o)\}$	✗ (not all (x, x) exist)	✗ (l, n exists, but n, l does not)	✗ (l, n and n, l exist together)	✗ (l, n and n, o exist, but l, o is not necessarily in the relation)

2. Let $R_1 = \{(8, j), (j, k), (k, 5)\}$ and $R_2 = \{(8, 8), (8, j), (j, 8), (j, j), (j, k), (k, 8), (k, j), (k, k), (k, 5)\}$ be relations from $\{8, j, k\}$ to $\{8, j, k, 5\}$. Find

a) $R_1 \cup R_2$

b) $R_1 - R_2$

c) $R_1 \cap R_2$

d) $R_2 - R_1$

$R_1 \cup R_2: \{(8, 8), (8, j), (j, 8), (j, j), (j, k), (k, 8), (k, j), (k, k), (k, 5)\}$

$R_1 - R_2: \emptyset$

$R_1 \cap R_2: \{(8, j), (j, k), (k, 5)\}$

$R_2 - R_1: \{(8, 8), (j, 8), (j, j), (k, 8), (k, j), (k, k)\}$